Project Report: CSR Center Student & inventory Management Web Portal

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Aim and Scope

Build a local demonstration portal for CSR centre staff to maintain student records, record inventory receipts and issues, and export stock summaries. Project reference: PRJ-CP-401. All records and results in this report are fictional demo evidence.

Materials and Tools

One training-centre computer, Python 3, HTML/CSS, SQLite, a browser and a code editor. Used Python csv and sqlite3 modules for exports and storage; no paid software was required.

Design and Procedure

1. Defined student fields (code, name, trade, centre) and item fields (code, name, unit). Stored receipts and issues as transactions with quantity, date and reason.

2. Created SQLite tables with unique student/item codes and foreign keys. Used parameterised SQL and a transaction for each stock issue.

3. Built labelled HTML/CSS forms, search results and validation messages. Rejected blank required fields, duplicate codes and non-positive quantities.

4. Calculated available stock from receipts minus issues. Blocked an issue exceeding available stock and exported the stock summary as CSV for Excel.

5. Added staff sign-in, checked access on protected pages, and documented a database backup and restore exercise using synthetic records only.

Safety and Data Protection

Used fictional names and contact details; kept the SQLite file outside the public directory. Stored password hashes, kept credentials out of the report, and signed out on shared computers.

Limited write access to authorised staff. Kept a backup before changes and used safe posture and regular screen breaks during computer work.

Test Cases and Observations

Demo test set: 20 fictional students, 15 inventory items and 30 stock transactions. Ran 12 checks: 4 student validation, 4 stock validation, 2 access and 2 export/restore checks.

First pass: 10/12 checks passed. Fixed acceptance of a negative issue quantity and a missing duplicate-code message. Re-ran all 12 checks: 12/12 passed in the demo scenario.

Stock reconciliation example: opening quantity 10 keyboards + receipt 5 - issue 3 = 12 available. An attempted issue of 13 was rejected and the balance stayed at 12.

Restored the backup into a separate database and checked all 20 student records and 15 item balances against the original. CSV row counts and stock totals matched.

Results and Limitations

Delivered working student forms, inventory transactions, stock CSV export, a test checklist and a short staff guide in this fictional exercise. The prototype was tested locally; multi-user load, production hosting and independent security review remain future work.

Logbook and Bill of Materials

24 hours over 4 days: requirements and schema 4 h; forms and database 8 h; validation and testing 8 h; documentation and handover 4 h.

BOM: existing centre computer and internet, Python, SQLite and editor: Rs. 0 additional procurement. Hardware and connectivity were provided by the centre.

Conclusion

The exercise connected Python, Web Design and Excel Automation with Software QA. Validation and stock reconciliation made the demonstration more reliable. Next steps are role-based access review and testing simultaneous stock updates.